

Traffic Intelligence Report

Exploratory Data Analysis & Strategic Insights for Urban Mobility

Executive Overview

Dataset Snapshot

An extensive analysis of urban traffic flows spanning 48,024 chronological records across 4 key city junctions.

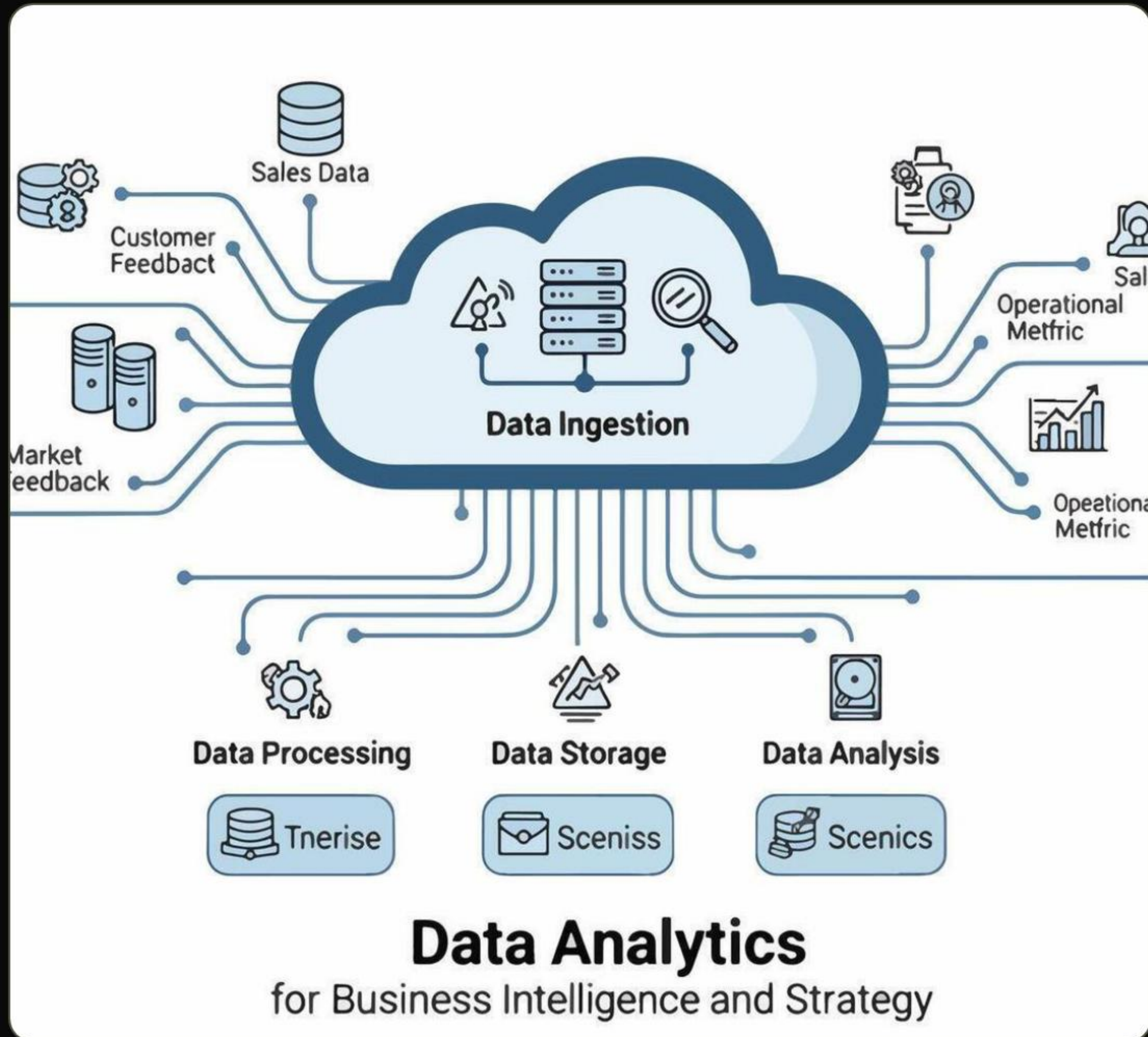
- ✓ 48k+ Time-series observations
- ✓ 17+ Engineered features
- ✓ Multimodal signal integration



Core Objective

To identify the primary drivers of traffic volume and prepare a high-fidelity feature matrix for predictive modeling.

Pipeline Architecture



01. Preprocessing

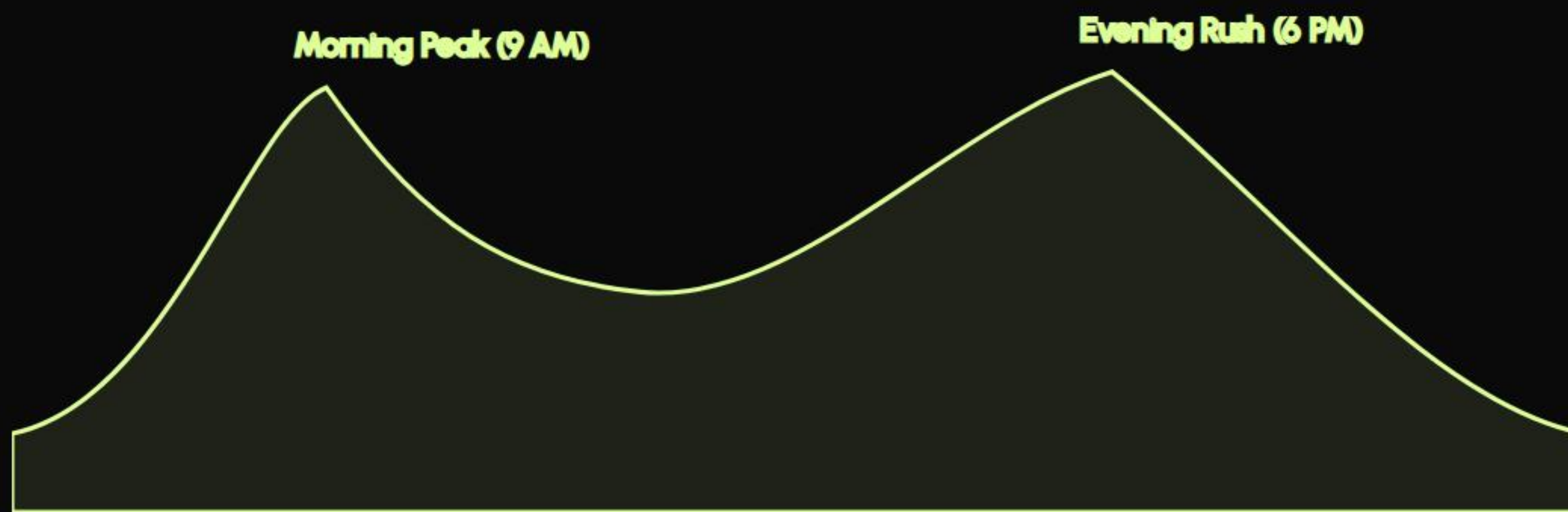
Standardizing continuous environmental variables and cleaning chronological sequences.

02. Feature Engineering

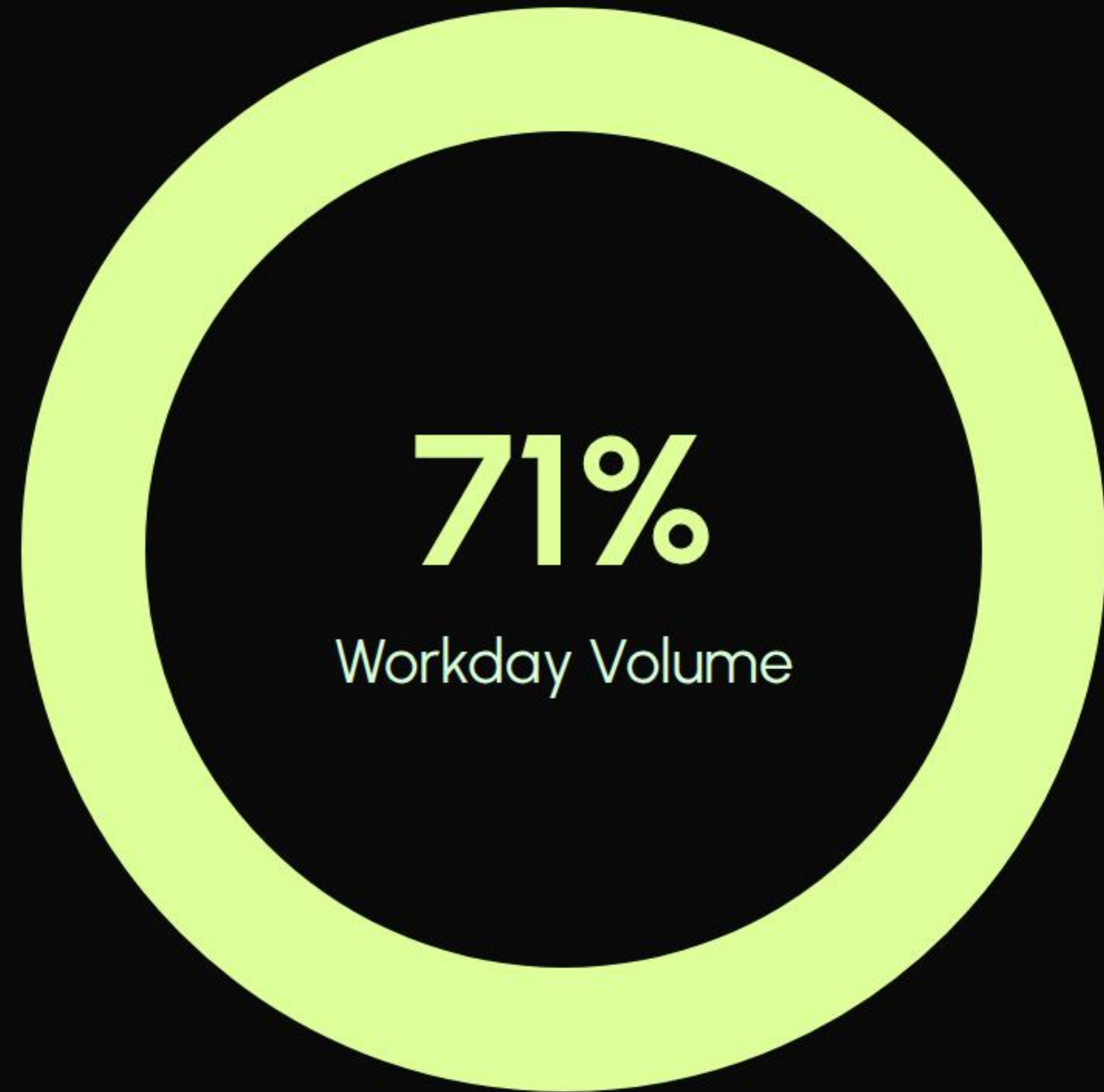
Generating temporal lags and behavioral flags to capture sequential dependencies.

The **Pulse** of Mobility

Analysis reveals a distinct bimodal peak pattern, characterizing the urban commuter rhythm.



Weekly Seasonality



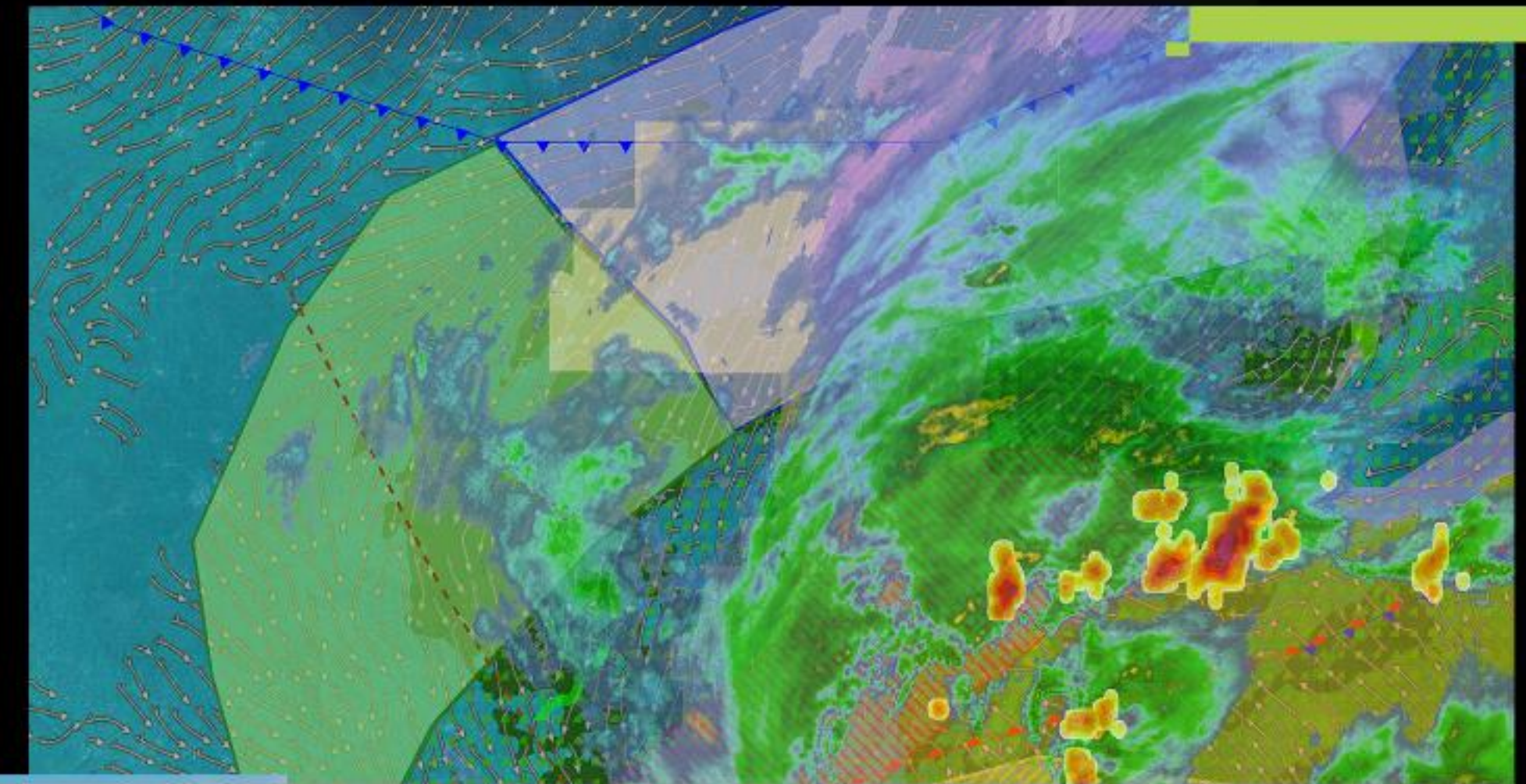
- **Workdays:** Highly predictable commuter peaks.
- **Weekends:** Smoother, leisure-driven distribution.
- **Holidays:** Significant anomalies in volume.

Weather Dynamics

While weather parameters like temperature and wind speed were measured, EDA shows they act as **minor behavioral modifiers** rather than primary drivers.

Standardization Note

Continuous features like *temperature_2m* and *relative_humidity_2m* were Z-scaled to mean 0, variance 1.



Sequential Lags



Lag_1

Traffic count from 1 hour ago.
Captures the most immediate temporal dependency.



Lag_2

Traffic count from 2 hours ago.
Helps in identifying short-term velocity trends.



Lag_24

Traffic from same hour yesterday.
Captures daily cyclic seasonality.

The Dominance Factor



Key Finding: Immediate history (lag_1) is the single most powerful predictor, representing over 93% of global split importance.

Mutual Information Gain

Validating non-linear signals. MI highlights structural features often overlooked by tree models.

Feature	MI Score	Classification
lag_1	1.2144	Dominant Signal
junction	0.3994	Critical Context
hour_only	0.0925	Cyclic Signal
is_concert	0.0046	Event Trigger

Anomaly Intelligence



Event Overrides

Special events like concerts and sports events show low *Global* importance but high *Local* impact.

- **Protecting Accuracy:** These flags prevent the model from treating surges as unexplainable errors.
- **Sparse Signals:** Even with low frequency, they are vital for edge-case forecasting.

Optimized **Feature** Set



Dropped Features

Removed to reduce noise and computational complexity:

- Rain (Redundant with Temp)
- Wind Speed (Low MI/Importance)
- Snowfall (Zero Variance)



Retained Features

Critical structural and temporal drivers:

- Lags (1, 2, 24)
- Spatial (Junction)
- Event Flags (Concerts/Sports)

Strategic Roadmap

Predictive Strategy

With a pruned, high-fidelity feature matrix, the next phase focuses on **XGBoost Regressor** implementation for short-term forecasting.

"We have transformed raw environmental data into actionable mobility intelligence."



Image Sources

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